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Adapter for patient reference array and bone anchor assembly

Abstract

An apparatus includes a reference array, a bone anchor, a receiver assembly coupled to the bone anchor, and an adapter axially locked with the receiver assembly. The adapter includes a proximal end coupled to the reference array and a distal end coupled to the receiver assembly.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/294,229, filed Dec. 28, 2021, which is incorporated herein by reference in its entirety for any and all purposes.

BACKGROUND

- (1) Spinal surgical procedures often include application of bone anchor assemblies (e.g., bone screw assemblies) that are connected by rigid spinal rods locked to each bone anchor assembly. The bone anchor assemblies can often include a bone anchor and a rod-housing component, which can be referred to as a “receiver” or “tulip”, that is often coupled to the bone anchor in a manner that permits angular adjustability of the tulip relative to the bone anchor.
- (2) A noteworthy trend in the medical community is the move away from performing surgical procedures on the spine via traditional “open” techniques in favor of so-called “minimally invasive” or “minimal access” techniques. Open surgical techniques are generally undesirable in that they typically require large incisions with high amounts of tissue displacement to gain access to the surgical target site, which produces concomitantly high amounts of pain, lengthened hospitalization (increasing health care costs), and high morbidity in the patient population. Less-invasive surgical techniques (e.g., including minimal access and minimally invasive techniques) are gaining favor because they involve accessing the surgical target site via incisions of substantially smaller size with greatly reduced tissue displacement requirements. This, in turn, reduces the pain, morbidity, and cost associated with such procedures.
- (3) One disadvantage to performing minimally invasive surgery is the increased reliance on radiographic imaging to “see” the spinal target site, instruments, and implants during the surgery. While this increased exposure is generally negligible for the patient, over time and over multiple procedures on different patients, this increased exposure adds up for the surgeon and other operating room personnel. Systems and methods have been developed to reduce reliance on radiographic imaging during spine surgery. Once such system and method involve three-dimensional (3D) navigation systems that use positional tracking systems to track the position of implants and instruments relative to the spinal target site and present the surgeon with a representative image of the instrument superimposed on an image of the target site to indicate the position of the implant or instrument relative to the anatomical structures depicted in the image. (e.g. spinal target site). However, these systems have the disadvantages of generally requiring that reference markers be somehow fixed to the patient (e.g. anchoring a pin or other instrument into the patient's spine, thus causing additional trauma to the patient's anatomy), requiring input of pre-operative CT images into the system before or during the procedure, and/or requiring large, specialized, and expensive equipment that may not be available in certain instances or operating rooms. Furthermore, even though 3D navigation provides spatial information regarding a target surgical site, instruments, and implants during surgery, it does not provide neurophysiologic information regarding the nerves lying near and around the operative corridor.
- (4) There is a need for systems, methods, and apparatuses that address one or more shortcomings in

the art and provide one or more advantages.

SUMMARY

(5) A first example apparatus includes: a reference array, a bone anchor, a receiver assembly coupled to the bone anchor, and an adapter axially locked with the receiver assembly. The adapter includes a proximal end coupled to the reference array and a distal end coupled to the receiver assembly. The receiver assembly can include one or more tool engagers. The adapter can include one or more receiver assembly engagers coupled with the one or more tool engagers. The one or more receiver assembly engagers can include one or more arms. The receiver assembly can include at least two receiver arms defining a channel therebetween configured to receive a spinal rod. The adapter can include a channel engager configured to engage within the channel. The adapter can include a driver configured to advance the channel engager into the channel and withdraw the channel engager from the channel.

(6) A second example apparatus includes a proximal end having a reference array coupler, a distal end defining an opening configured to receive a receiver assembly, and one or more receiver assembly engagers configured to engage with the receiver assembly.

(7) The one or more receiver assembly engagers can include one or more arms biased into a locked position. The one or more arms can include one or more tabs configured to fit within one or more tool engagers of the receiver assembly. The channel engager can be configured to engage within a channel formed by the receiver assembly. The channel engager can include a channel engager shaft having a channel engager bar extending perpendicular to a length of the channel engager shaft. The channel engager bar can be the portion of the channel engager that engages within the channel. The channel engager bar can mimic a shape and diameter of a spinal rod. The driver can be configured to advance the channel engager into the channel and withdraw the channel engager from the channel. The driver can be configured as a lead screw configured to translate rotational force applied to the driver to advance or withdraw the channel engager.

(8) An example method can include: coupling an adapter with a receiver assembly, coupling a reference array to a proximal end of the adapter, and adjusting an angle of a receiver assembly with respect to a bone anchor. The adjusting can simultaneously adjust an angle of the adapter with respect to the bone anchor.

(9) The coupling of the adapter with the receiver assembly can include disposing one or more tabs of an arm of the adapter within one or more tool engagers of the receiver assembly. The method can further include, after adjusting the angle, advancing a channel engager of the adapter into a channel of the receiver assembly, thereby locking an angle of the receiver assembly with respect to the bone anchor. The method can further include advancing a channel engager of the adapter toward a channel of the receiver assembly such that the channel engager contacts a set screw or rod disposed in the receiver assembly. The method can be performed during a revision spinal surgery. The bone anchor may have been placed during a prior spinal surgery. The method can further include uncoupling the adapter from the receiver assembly and connecting the receiver assembly with another receiver assembly via a rod.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates, with respect to a patient's spine, an apparatus that includes a reference array coupled to a bone anchor assembly via an adapter.

(2) FIG. 2 illustrates a perspective view of the adapter.

(3) FIG. 3 illustrates a front view of the adapter.

(4) FIG. 4 illustrates a side view of the adapter.

(5) FIG. 5 illustrates a top view of the adapter.

- (6) FIG. **6** illustrates a bottom view of the adapter.
- (7) FIG. **7** illustrates an exploded view of the adapter.
- (8) FIG. **8** illustrates a cutaway view of the adapter.
- (9) FIG. **9** illustrates an example method.
- (10) FIG. **10** illustrates a cutaway view of the adapter coupled to the bone anchor assembly.
- (11) FIG. **11** illustrates a cutaway view of the reference array coupled to the adapter.
- (12) FIG. **12** illustrates a cone of angulation of an adapter coupled to a bone anchor assembly.
- (13) FIG. **13** illustrates a view of the adapter coupled to the bone anchor assembly with a channel engager bar deployed into a channel of the bone anchor assembly.
- (14) FIG. **14** illustrates the adapter coupled to a receiver assembly of a construct lacking a set screw.
- (15) FIG. **15** illustrates the adapter coupled to a receiver assembly of a construct having a set screw.
- (16) FIG. **16** illustrates an exploded view of an example bone anchor assembly.
- (17) FIG. **17** illustrates an example system.

DETAILED DESCRIPTION

(18) An example adapter is configured for attaching a reference array to a bone anchor assembly implanted in a patient. The attachment of the adapter facilitates the customizable directionality of the reference array during a navigated procedure, such as a spine procedure. Bone anchor assemblies used in spine procedures can include a component known as a “tulip” or “receiver”, which is a component configured to receive and anchor a spinal rod. The adapter includes a distal end that is configured to connect with the tulip via two arms that fit into tool engagers of the tulip. The adapter can be coupled such that long axes of the adapter and the tulip are coaxial or at least parallel. Once connected, the reference array is pivotable about a head of a bone anchor to which the tulip is connected and with which the tulip has a polyaxial relationship. After pivoting the reference array (e.g., such that the reference array faces a camera of a surgical navigation system, an inner shaft of the adapter is driven into the tulip in a manner that locks the tulip head in place and creates a rigid construct.

(19) In an example, the adapter is configured to couple with a single bone anchor assembly. In an example, the adapter is configured to couple with multiple bone anchor assemblies. In an example, the adapter lacks a feature configured to couple to a spinal rod.

(20) An example implementation of an adapter with respect to a patient's spine is shown and described in relation to FIG. **1**.

(21) FIG. **1** illustrates, with respect to a patient's spine, an apparatus **10** that includes a reference array **20** coupled to an adapter **100** that is coupled to a bone anchor assembly **50**.

(22) The illustrated bone anchor assembly **50** includes a receiver assembly **70** coupled to a bone anchor **60** implanted in a pedicle of vertebra of the spine. The bone anchor assembly **50** can be configured the receiver assembly **70** is couplable to a polyaxial bone anchor (e.g., a pedicle screw). The receiver assembly **70** can be initially movable relative to the bone anchor in a polyaxial fashion and subsequently lockable relative to the bone anchor (e.g., prior to or as part of locking a rod in the receiver assembly **70**). Additional details regarding an example implementation of a bone anchor assembly **50** are provided in relation to FIG. **16**. Bone anchor assemblies **50** can take any of a variety of forms and are commonly used in spinal procedures. Many implementations include a receiver assembly, collet, and bone anchor. Example implementations are described in US 2020/0367940, filed May 22, 2019, and U.S. Pat. No. 8,986,349, filed Nov. 12, 2010, both of which are incorporated herein by reference in their entirety for any and all purposes.

(23) The illustrated reference array **20** includes one or more fiducials **30**. The body of the reference array supports the one or more fiducials **30**. The spacing, orientation, and other ways in which the fiducials **30** are disposed at the reference array can be used to convey meaning to a user or surgical navigation system. In the illustrated example, the fiducials **30** are passive infrared retroreflectors configured to reflect infrared light emitted by a tracking system back to the tracking system. In

other examples, the fiducials **30** are active infrared emitters that actively emit infrared light. In still other examples, the fiducials **30** are some other object meaningful to implementing surgical navigation, such as the receiver or transmitter of energy used in tracking (e.g., ultrasonic energy or electromagnetic energy) or a visible light tracking glyph. In some examples, the reference array **20** includes a mix of types of fiducials (e.g., a mix of fiducials **30** configured to act in visible and infrared light).

(24) The illustrated reference array **20** further includes a fastener **22** for coupling the reference array with the adapter **100**. As illustrated, the fastener **22** is a threaded fastener that mates with threads of the adapter **100**, but the fastener **22** can take any of a variety of forms, such as a snap fit fastener, bayonet coupler, a magnetic fastener, other kinds of couplers, or combinations thereof.

(25) The adapter **100** is a component configured to facilitate coupling of the reference array **20** to a bone anchor assembly **50** and is described in more detail in figures herein.

(26) FIGS. **2-8** illustrate various views of an example implementation of the adapter **100**. In particular, FIG. **2** illustrates a perspective view of the adapter **100**. FIG. **3** illustrates a front view of the adapter **100**. FIG. **4** illustrates a side view of the adapter **100**. FIG. **5** illustrates a top view of the adapter **100**. FIG. **6** illustrates a bottom view of the adapter **100**. FIG. **7** illustrates an exploded view of the adapter **100**. FIG. **8** illustrates a cutaway view of the adapter **100**.

(27) The adapter **100** includes a body **101** defining proximal end **102** and a distal end **104**. At the proximal end **102**, the body **101** defines a reference array coupler **108**. At the distal end, the body **101** defines an opening **106**. Coupled to the body **101** is at least one receiver assembly engager **110**. The adapter **100** further includes a channel engager **150** that is driven by a driver **160**.

(28) The opening **106** can be sized and shaped to fit at least a portion of the receiver assembly **70**. The portion of the receiver assembly **70** (e.g., a proximal end of the receiver assembly **70**) can fit within the opening **106**. The fit can be sufficiently tight as to permit relatively easy insertion of the receiver assembly **70** by a user while resisting lateral movement or tipping of the body **101** relative to the receiver assembly **70** once the receiver assembly **70** is sufficiently inserted within the opening **106**. An example of the fit is shown in FIG. **10**, which is discussed in more detail infra.

(29) The reference array coupler **108** is a portion of the adapter **100** configured to couple with a reference array **20**. The reference array coupler **108** can couple via any of a variety of kinds of mechanisms. For example, the illustrated example of the reference array coupler **108** has threads for coupling with threads of the screw fastener **22** for securing the reference array **20**. In other implementations, the reference array coupler **108** can have other kinds of mechanisms for coupling, such as a snap fit fastener, bayonet coupler, a magnetic fastener, other kinds of couplers, or combinations thereof. In addition, the overall shape of the reference array coupler **108** can facilitate coupling with the reference array **20**. As illustrated, the reference array coupler **108** can fit within a portion of the reference array **20** to resist movement of the reference array **20** with respect to the reference array coupler **108** in at least one direction.

(30) In the illustrated example, the reference array coupler **108** is in a fixed relationship with the body **101**, such that the angle of the reference array coupler **108** or a reference array **20** coupled thereto cannot be adjusted relative to the body **101**. In other examples, the reference array coupler **108** or the reference array **20** coupled thereto is adjustable relative to the body **101**. For example, the body **101** may define a central axis along the length of the body (e.g., from proximal end **102** to distal end **104**) and the angle along which the reference array **20** couples to the coupler **108** can be adjusted relative to the central axis. Various adjustment mechanisms can be used, such as via one or more articulating joints.

(31) The receiver assembly engager **110** is a mechanism of the adapter **100** configured to couple with a receiver assembly **70**. The arrangement of the receiver assembly engager **110** can vary depending on the arrangement of the receiver assembly **70**. In an example, the receiver assembly **70** includes a tool engager (e.g., a groove, detent, or other feature configured to couple with a rod inserter or other instrument) and the receiver assembly engager **110** is configured to couple thereto.

An example implementation is illustrated.

(32) In the illustrated example, the adapter **100** has two receiver assembly engagers **110**. Each receiver assembly engager includes an arm **210** having an arm interface **214** at a proximal end and a tab **216** at a distal end. The arm **210** further includes an arm pivot **212** and the arm **210** engages with an arm spring **218**.

(33) The arm **210** is an elongate structure that supports or defines components useful in engaging with the receiver assembly **70**. As illustrated, the arm **210** runs along an exterior of the adapter **100**.

(34) The arm interface **214** is a portion of the arm **210** configured to receive interaction from the user. For example, the arm interface **214** can be in the form of an area for the user to press to cause the arm **210** to rotate about the arm pivot **212** to cause the distal end of the arm **210** to move outward. This movement can be used to decouple the receiver assembly engager **110** from the receiver assembly **70**. The arm interface **214** includes grooves or other features (e.g., knurling or coloring) to facilitate the interaction with the user.

(35) The tab **216** is a portion configured to interact with the tool engager of the receiver assembly **70**. In the illustrated example, the tab **216** extends inward from the arm **210** to fit within the tool engager of the receiver assembly and resist proximal movement of the adapter **100** relative to the receiver assembly **70**. The tab **216** can extend from the arm **210** such that the tab **216** extends into the opening **106** to reach the receiver assembly **70** when the receiver assembly **70** is received within the opening **106**. The tab **216** can include a ramped distal surface such that when the tab **216** is pushed distally onto the receiver assembly **70**, the interaction therebetween causes the distal end of the arm **210** to be pushed outward. Continued pushing can result in the tab **216** falling (or being pushed) into the tool engager of the receiver assembly **70**. A proximal end of the tab **216** can be flat, such that the proximal end of the tab **216** catches on the tool engager and resists proximal movement of the adapter **100** relative to the receiver assembly **70**. Further details regarding this interaction is described in more detail in relation to FIG. **10**.

(36) In the illustrated example, the arms **210** end in tabs **216** and the tabs **216** are configured to interact with the tool engagers. In other examples, the arms **210** can include other components in addition to or instead of the tabs **216**. The configuration of such components can vary depending on the kind of receiver assemblies **70** to which the adapter **100** is to be coupled.

(37) The arm pivot **212** is a pivot portion about which the arm **210** pivots relative to the body **101**. The arm pivot **212** can be coupled to the body **101** via an arm pin **220**. In the illustrated example, the arm pivot **212** is an extended piece of material having a passage extending therethrough that accommodates an arm pin **220**. In other implementations, the arm pivot **212** can be or include one or more features configured to interact with the body to facilitate pivoting or otherwise moving the arm **210** with respect to the body **101**.

(38) The arm spring **218** is a component of the adapter **100** that biases the arm **210** into a closed position. As illustrated, the arm spring **218** is disposed toward the proximal end of the arm **210** and pushes the proximal end outward such that the distal end of the arm **210** is urged inward. Such urging inward can encourage the tab **216** to fall into the tool engager and resist unwanted disengagement. In other implementations, in addition to or instead of an arm spring **218**, other biasers can be used.

(39) The channel engager **150** is a portion of the adapter **100** configured to engage with a channel of the receiver assembly **70**. The illustrated channel engager **150** includes an elongate channel engager shaft **152** having, at its distal end, a channel engager bar **154**. As illustrated, the channel engager shaft **152** is an elongate structure running parallel to and coaxial with the length of the body **101**. As illustrated, the channel engager shaft **152** has a length less than a length of the body, but can have other lengths in other implementations. A channel engager pin **156** is coupled to the channel engager **150** and facilitates guiding the channel engager down a track in the body **101**.

(40) The illustrated channel engager bar **154** is an end effector of channel engager **150** that is configured to engage with the channel of the receiver assembly **70**. In this example, the end

effector is in the form of a channel engager bar **154**. The bar **154** has a length that extends perpendicularly from a distal end of the channel engager shaft **152**. As illustrated, the length of the bar **154** is not greater than a width of the body **101** (e.g., the width can be greatest width of the body **101** in a direction parallel to the length of the bar **154**). The bar **154** can be sized and shaped to mimic a spinal rod that would be inserted into a channel of the receiver assembly. In addition or instead, the bar **154** can have a flat distal surface to facilitate interaction with a set screw, other flat feature of the receiver assembly **70**, or a non-circular rod. In other examples, the bar **154** can have a convex distal surface to facilitate engagement between the bar **154** and a spinal rod already inserted in the receiver assembly. In many examples, the channel engager bar **154** is formed from a rigid material. In some examples, the channel engager bar is constructed from or includes a pliable material configured to facilitate grip between the channel engager **150** and the receiver assembly **70**.

(41) In other examples, the end effector of the channel engager **150** is not the illustrated bar and is a threaded component that mimics the size of a set screw that would be disposed in the receiver assembly **70** to, for example, lock a rod. As described elsewhere herein, the receiver assembly **70** can include threads or grooves configured to interface with a set screw. The end effector of the channel engager **150** can be configured to thread into these threads or grooves to interact with the receiver assembly.

(42) The channel engager **150** can include or be configured to interact with (e.g., via a feature on a proximal end of the channel engager) a driver **160**.

(43) The driver **160** is a component configured to facilitate the advancement and retraction of the channel engager. In the illustrated example, the driver **160** is arranged as a leadscrew mechanism that pushes against the channel engager **150** as a feature of the driver **160** is advanced by a screwdriver. The driver **160** can be arranged with a distal feature that pulls on the channel engager **150** as the feature of the driver is retracted with a screwdriver. The proximal end of the driver **160** can include the feature that interacts with a screwdriver and the feature can be accessible through the proximal end of the adapter **100**.

(44) The adapter **100** can be used during a surgical procedure to facilitate navigation with the reference array **20**.

(45) FIG. 9 illustrates an example method **900** of using the adapter **100**. The method **900** can be performed during a surgery. In an example, the surgery can be a spinal surgery, such as a revision spinal surgery. The method **900** can include placing a bone anchor **60** or exposing a bone anchor **60** that was placed during a prior spinal surgery. In the illustrated example, the method **900** begins with operation **910**.

(46) Operation **910** includes coupling the adapter **100** with the receiver assembly **70**. In an example, the operation **910** occurs while the receiver assembly **70** is coupled to bone anchor **60** that is anchored in bone. For example, the bone anchor **60** can be a bone screw anchored in a patient's pedicle, and the receiver assembly **70** is configured to receive a spinal rod. The operation **910** can be accomplished in any of a variety of ways depending on how the adapter **100** and receiver assembly **70** are arranged, such as via snapping, twisting, threading, ratcheting, other fastening actions, or combinations thereof. In an example, the coupling of the adapter **100** with the receiver assembly **70** includes disposing one or more tabs **216** of the one or more arms **210** of the adapter **100** within one or more tool engagers **72** of the receiver assembly **70**. For instance, the opening **106** at the distal end **104** of the adapter **100** located over the receiver assembly **70** and pushed downward. As the adapter **100** is pushed downward, a ramped distal surface of the one or more tabs **216** contacts the proximal end of the receiver assembly **70** (e.g., a proximal end of the receiver arms **74**) and causes the distal end of the arm **210** to be pushed outward. The user continues pushing the adapter **100** onto the receiver assembly **70**, which results in the tab **216** falling into the tool engager of the receiver assembly **70**, which results in the arrangement shown in FIG. 10. The tabs **216** can be inward facing such that they extend toward a central axis of the adapter **100**.

(47) FIG. 10 illustrates a cross-sectional view an example coupling between the adapter **100** and the receiver assembly **70**. Here, the tabs **216** of the arms **210** have fallen into tool engagers **72** of receiver arms **74** of the receiver assembly **70**. As illustrated, the tool engagers **72** take the form of grooves in the receiver arms **74**. While in this arrangement, proximal movement of the adapter **100** would be resisted by an interaction between the proximal flat surface of the tabs **216** and the proximal flat surfaces of the tool engagers **72**. As can be further seen in this view, the fit between the opening of the adapter and the outer surface of the receiver assembly is sufficiently tight as to resist significant movement relative to the receiver assembly (e.g., movement that would cause accuracy errors outside of a tolerated range for a navigation system relying on a reference array **20** held by the adapter **100**). As further illustrated, the bar **154** is in a retracted (e.g., undeployed) state, and a collet **78** of the bone anchor assembly **50** is in such a position as to not resist polyaxial movement of the receiver assembly **70** relative to the bone anchor **60**. Additionally, the illustrated adapter **100** is advanced as far as possible onto the receiver **70** because a portion of the body **101** forming a portion of the proximal end of the opening **106** abuts the receiver **70** and prevents further advancement. The figure further shows that distal movement of the collet **78** (e.g., as a result of pressure from the bar **154**) can cause the collet **78** to interact with the bone anchor **60** and the receiver assembly **70** so as to resist polyaxial movement of the receiver assembly **70** relative to the bone anchor **60**.

(48) Returning to FIG. 9, operation **920** includes coupling a reference array **20** to a proximal end of the adapter **100**. The specific coupling method can vary depending on the specifics of the reference array **20** and the reference array coupler **108**. The coupling can include snapping, twisting, threading, ratcheting, other fastening actions, or combinations thereof to make the connection. An example connection between the reference array **20** and the reference array coupler **108** is shown in FIG. 11.

(49) FIG. 11. illustrates a cross section view of a reference array **20** coupled to a reference array coupler **108**. The illustrated connection is made via a screw fastener **22** that mates with corresponding screws of the reference array coupler **108**. In addition, the figure illustrates that the proximal end of the adapter **100** can at least partially fit within the reference array **20**. Such a fitting can resist the reference array from rotating with respect to the adapter **100**. The illustrated reference array **20** has a width and length extending perpendicular to a long axis of the adapter **100** while the thickness of the reference array **20** is parallel to the adapter **100** when the reference array **20** is coupled to the adapter **100**. Likewise, based on the connection between the adapter **100** and the receiver assembly **70**, the width and length of the reference array **20** can extend perpendicular to a long axis of the receiver assembly **70** (e.g., the axis along which the arms of the receiver assembly **70** extend) and the thickness of the reference array **20** is parallel to the long axis of the receiver assembly **70** when the reference array **20** is coupled to the adapter **100** and the adapter **100** is coupled to the receiver assembly **70**.

(50) Returning to FIG. 9, operation **930** includes adjusting an angle of the receiver assembly **70** with respect to the bone anchor **60**. The adjusting can simultaneously adjust an angle of the adapter **100** with respect to the bone anchor **60**. The adjusting can be to move the reference array **20** from a position better for use during surgical navigation (e.g., because the reference array **20** is more visible to a navigation camera system or less likely to be obscured during a procedure). The adjusting can include moving the reference array **20**, which causes the adapter **100** to move due to the rigid connection with the reference array **20**, which causes the receiver assembly **70** to move relative to the bone anchor **60** due to their movable connection. The adjusting can include one, two, or three of pitching, rolling, and yawing the reference array with respect to the bone anchor **60**.

(51) FIG. 12 illustrates an example cone of angulation **1200** for the apparatus **10**. The cone of angulation **1200** can represent an amount of movement of the adapter **100** afforded by the connection between the adapter **100** and the receiver assembly **70**, which has a movable relationship with the bone anchor **60**.

(52) Returning to FIG. 9, operation **940** includes advancing a channel engager **150** of the adapter **100** into a channel **76** of the receiver assembly **70**, thereby locking an angle of the receiver assembly **70** with respect to the bone anchor **60**. The technique for advancing can vary depending on the configuration of the channel engager **150**. In the examples illustrated above, the channel engager **150** is able to be advanced or retracted via a driver **160** that pushes or pulls on the channel engager **150**. In the illustrated configuration, the driver **160** acts as a leadscrew that transfers rotation energy imparted at a proximal end of the driver (e.g., via a screwdriver) into linear motion that moves the driver and causes the channel engager **150** to move. Advancing the channel engager **150** can include causing the driver **160** to push the channel engager shaft **152** such that the channel engager bar **154** (or other engagement element) advances into the channel **76** and contacts one or more components of the bone anchor assembly **50**. The advancing can be configured to achieve or further strengthen a rigidity of a connection between the receiver assembly **70** and the adapter **100**. For instance, the advancing can cause the channel engager **150** (e.g., the bar **154** thereof) to contact a component of the bone anchor assembly **50** and urge the adapter **100** upward (e.g., proximally). The upward urging can cause a proximal end of a flat of the one or more tabs **216** to abut a proximal end of the tool engager **72** and increase a rigidity of the connection by taking up play in the connection. In addition or instead, the advancing of the channel engager can cause resistance in the movement of the receiver assembly **70** relative to the bone anchor **60**. For instance, the advancing of the channel engager **150** can push the collet **78** downward (distally) such that the relationship between the collet **78** and the receiver assembly **70** and the bone anchor **60** resists movement of the receiver assembly **70** relative to the bone anchor **60**. Examples of the channel engager **150** in an advanced state are shown in FIGS. **13-15**.

(53) FIG. **13** illustrates an example channel engager **150** advanced into the channel **76** of the receiver assembly **70**. Here, the channel engager **150** is in contact with the collet **78** after the channel engager drives the collet **78** into such a position that the collet **78** resists movement of the receiver assembly **70** relative to the bone anchor **60**.

(54) FIG. **14** illustrates an example channel engager **150** advanced toward a bone anchor assembly **50** of a construct **52** that includes multiple bone anchor assemblies **50** linked by a rod **80**. The rod **80** is disposed in channels **76** of multiple bone anchor assemblies **50** and held in place by set screws **79**. Here, the channel engager **150** is sufficiently advanced that the bar **154** is in contact with a set screw **79**. Depending on the arrangement of the bone anchor assemblies **50**, the set screw **79** may cause the rod **80** to advance collets to fix the relationship between the receiver assemblies **70** and the bone anchors **60**. In an example, the construct **52** is a preexisting construct from a prior surgery and the current surgery is a revision surgery.

(55) FIG. **15** illustrates an example channel engager **150** in contact with a rod **80** of a construct **52** that includes multiple bone anchor assemblies **50** linked by the rod **80**. The construct **52** is similar to the one in FIG. **14** except, here, the bar **154** is directly in contact with the rod **80** rather than a set screw.

(56) Returning to FIG. **9**, operation **950** includes uncoupling the adapter **100** from the receiver assembly **70**. How the uncoupling is accomplished can vary depending on how the adapter **100** was coupled to the assembly **70**. In an example, the uncoupling includes retracting the channel engager **150**, such as by driving the driver **160** in reverse to cause the driver **160** to pull the channel engager **150** distally. Depending on the arrangement of the adapter, the receiver assembly **70** may need to be removed to access the driver **160**. In an example, the uncoupling includes pressing in on the one or more arm interfaces **214** to cause the distal ends of the arms **210** to swing outward causing the tabs **216** to withdraw from the tool engagers **72** of the receiver assembly **70**. While the tabs **216** are out of the tool engagers **72**, a user can pull the adapter **100** upward to free the receiver assembly **70** from the opening **106**.

(57) Operation **960** can include connecting the receiver assembly **70** with another receiver assembly **70** via a rod **80**. For example, the operation **960** can include implanting another bone

anchor assembly **50** having the another receiver assembly **70** (e.g., using the benefit of surgical navigation provided by the array **20**). After the implanting, the rod **80** can be disposed in the channels **76** of the two or more receiver assemblies **70** to be connected. The rod **80** can then be tightened in place by a set screw **79**. The resulting construct **52** can be similar to the construct shown in FIG. **14**.

(58) FIG. **16** illustrates an exploded view of an example bone anchor assembly **50**. The figure and its description are based on U.S. Pat. No. 8,986,349, which was previously incorporated herein by reference. Other implementations of bone anchor assemblies **50** are possible.

(59) The bone anchor assembly **50** includes a bone anchor **60** and a receiver assembly **70**. The bone anchor assembly **50** can be constructed from a suitable material, such as one or more metals (e.g. titanium or stainless steel).

(60) The bone anchor **60** is configured to attach securely within a bony structure (e.g. pedicle of a vertebra) and to allow the receiver assembly **70** to provisionally lock into position relative to the bone anchor **60** after placement of the bone anchor **60** within a bony structure. The receiver assembly **70** and bone anchor **60** are configured to engage with full polyaxial motion. The receiver assembly **70** and bone anchor **60** can also be provisionally locked (e.g., fixed relative to each other prior to final capture and locking of a spinal rod into the receiver). This versatile engagement between the receiver assembly **70** and bone anchor **60** provides both the ease of positioning and rod placement associated with polyaxial screws and the ability to leverage the bone anchor **60** to manipulate the vertebral body.

(61) The bone anchor **60** of the bone anchor assembly **50** includes a shank **67**, a body, and a capture structure **66**. At least one helically-wound bone implantable thread **68** extends radially from the body and functions to secure the placement of the bone anchor **60** within a bony structure. The capture structure **66** includes at least one tool engaging feature (e.g., one or more indentations configured to receive a screwdriver) that can be used, for example, to engage and attach various tooling for aligning and advancing the bone anchor **60** into a bony structure. The generally spherical shape of the capture structure **66** allows the capture structure **66** to articulate within the collet **78** to achieve the polyaxial motion between the bone anchor **60** and the receiver assembly **70**.

(62) The receiver assembly **70** is configured to receive an elongate structure (e.g. a rod) within a channel **76** and secure the rod within the channel **76**. Once the receiver assembly **70** and bone anchor **60** are securely oriented in the desired orientation and the rod is captured in the receiver assembly **70**, a set screw or another structure can be engaged to lock the rod in the receiver assembly **70**.

(63) The receiver assembly **70** is typically provided in an assembled state and includes a collet **78**. The receiver assembly has a generally U-shaped appearance with a generally cylindrical inner profile and a faceted outer profile. A base with a pair of upstanding receiver arms **74** form a U-shaped cradle which define the channel **76**. Receiver assemblies **70** may be provided in a variety of dimensions depending on the size and shape of the rod that it will be in secured frictional engagement with.

(64) The arms **74** typically have at least one helically-wound guide and advancement structure at least partially situated along their internal walls beginning from the top surface of the receiver arms **74**. The guide and advancement structure are configured to mate with at least one exterior helically-wound structure (e.g., a set screw).

(65) FIG. **17** illustrates an example system **1010** with which examples described herein can be used. This figure and its description are based on U.S. Pat. No. 9,510,771, filed Oct. 28, 2011, which is incorporated herein by reference in its entirety for any and all purposes. Examples described herein can be used with other kinds of systems.

(66) The illustrated surgical tracking system **1010** including a position tracking system **1012** and at least one surgical object to be tracked. Preferably, the position tracking system **1012** includes an infrared (IR) position sensor **1014** (also referred to as an “IR camera”), an IR-reflective tracking

array **1016** mounted on an intraoperative imaging system **1018**, an IR-reflective tracking array **1020** mountable to each surgical object to be tracked and a feedback and control device comprising a control unit **1022** and a display **1024**. The control unit **1022** has position tracking software and C-arm video import capabilities and is communicatively linked to the display **1024** so that information relevant to the surgical procedure may be conveyed to the user in a meaningful manner. By way of example, the relevant information includes, but is not limited to, spatial positioning data (e.g., translational data in the x, y, and z axes and orientation/rotational data) acquired by the IR position tracking sensor **1014**. The intraoperative imaging system **1018** may be any commercially available C-arm fluoroscope **1026** communicatively linked to a C-arm display **1028** with an IR-reflective tracking array **1016** attached, for example, to the signal receiver.

(67) Examples herein can facilitate monitoring the location and orientation of surgical access instruments which can aid in both the insertion and positioning of the surgical access instruments themselves, as well as aiding in the later insertion of instruments and/or implants through the surgical access instruments. In another example, tracking the location of surgical instruments within the surgical field may be accomplished with significantly decreased reliance on intraoperative imaging. In yet another example, the present invention may be used in conjunction with, or integrated into, a neuromonitoring system for assessing one or more of nerve proximity (and/or nerve directionality) and pedicle integrity, among other functions. In still another example, the present invention may facilitate safe and reproducible pedicle screw placement by monitoring the trajectory of various surgical instruments used during pilot hole formation and/or screw insertion. While the above examples are described in more detail below, it is expressly noted that they are set forth by way of example and that the present invention may be suitable for use in any number of additional surgical actions where tracking the 3D location of surgical instruments and implants within the surgical field and decreased exposure to x-ray radiation are desired.

Accordingly, it will be appreciated then that while the surgical tracking system is generally discussed herein as being attached to instruments such as dilating cannulas, tissue retractors, C-arms, pedicle access tools, etc., other instruments may be substituted depending on the surgical procedure being performed and/or the needs of the surgeon.

(68) Any of the features or attributes of the above described embodiments and variations can be used in combination with any of the other features and attributes of the above described embodiments and variations as desired. Various modifications, additions and other alternative embodiments are possible without departing from the true scope and spirit. The embodiments presented herein were chosen and described to provide an illustration of various principles of the present invention and its practical application to thereby enable one of ordinary skill in the art to utilize the invention in various embodiments and with various modifications as are suited to the particular use contemplated. All such modifications and variations are within the scope of the present invention as determined by the appended claims when interpreted in accordance with the benefit to which they are fairly, legally, and equitably entitled.

Claims

1. An apparatus comprising: a reference array; a bone anchor; a receiver assembly coupled to the bone anchor and including: at least two receiver arms defining a channel therebetween configured to receive a spinal rod; and one or more tool engagers; and an adapter axially locked with the receiver assembly having: a body, wherein the body defines a cavity; a channel engager slidably coupled to the body and housed within the cavity, wherein the channel engager is configured to engage within the channel; a channel engager pin removably coupled to the channel engager, the channel engager pin configured to engage a track in the body to facilitate guiding the channel engager down the track; one or more receiver assembly engagers pivotably coupled to the body and configured to releasably couple with the one or more tool engagers; a proximal end coupled to the

reference array; and a distal end coupled to the receiver assembly, wherein the one or more receiver assembly engagers are coupled to the body at the proximal end and configured to pivot when an arm of each of the one or more receiver assembly engagers engages an arm spring.

2. The apparatus of claim 1, wherein the channel engager includes: a channel engager shaft having a distal end; and a bar having a length that extends perpendicularly from the distal end of the channel engager shaft.

3. The apparatus of claim 2, wherein the bar defines a flat distal surface.

4. The apparatus of claim 2, wherein the bar defines a convex distal surface.

5. The apparatus of claim 2, wherein the bar mimics a shape and diameter of a spinal rod.

6. The apparatus of claim 1, wherein the one or more receiver assembly engagers each include: at least one tab extending toward a central axis of the adapter, the tab having a ramped distal surface.

7. The apparatus of claim 6, wherein the adapter includes a driver configured to advance the channel engager into the channel and withdraw the channel engager from the channel.

8. An apparatus comprising: a proximal end having a reference array coupler; a distal end defining an opening configured to receive a receiver assembly; a body, wherein the body defines a cavity; a channel engager slidably coupled to the body and housed within the cavity, wherein the channel engager is configured to engage within a channel formed by the receiver assembly; a channel engager pin removably coupled to the channel engager, the channel engager pin configured to engage a track in the body to facilitate guiding channel engager down the track; and one or more receiver assembly engagers pivotably coupled to the body and configured to releasably engage with the receiver assembly, wherein the one or more receiver assembly engagers are coupled to the body at the proximal end and configured to pivot when an arm of each of the one or more receiver assembly engagers engages an arm spring.

9. The apparatus of claim 8, wherein each arm is biased into a locked position; and wherein each arm includes one or more tabs configured to fit within one or more tool engagers of the receiver assembly.

10. The apparatus of claim 8, wherein the channel engager includes a channel engager shaft having a channel engager bar extending perpendicular to a length of the channel engager shaft; and wherein the channel engager bar is a portion of the channel engager that engages within the channel.

11. The apparatus of claim 8, further comprising: wherein a distal end of the channel engager bar has a flat or convex distal surface.

12. The apparatus of claim 11, wherein the channel engager bar mimics a shape and diameter of a spinal rod.

13. The apparatus of claim 10, further comprising: a driver configured to advance the channel engager into the channel and withdraw the channel engager from the channel.

14. The apparatus of claim 13, wherein the driver is configured as a lead screw configured to translate rotational force applied to the driver to advance or withdraw the channel engager.

15. A method comprising: coupling an adapter with a receiver assembly that is coupled to a bone anchor disposed in bone, wherein the coupling includes: engaging an interface of an arm pivotably coupled to a body of the adaptor such that the arm compresses a biasing member coupled to the body, wherein the arm is pivotably coupled to the body at a proximal end of the adaptor; releasing the interface to pivot the arm such that one or more tabs of the arm of the adapter are disposed within one or more tool engagers of the receiver assembly; coupling a reference array to a proximal end of the adapter; and adjusting an angle of the receiver assembly with respect to the bone anchor, wherein the adjusting simultaneously adjusts an angle of the adapter with respect to the bone anchor, wherein the adapter includes a channel engager slidably coupled to the body and housed within the cavity, wherein the channel engager is configured to engage within the channel, and wherein the adapter includes a channel engager pin removably coupled to the channel engager, the channel engager pin configured to engage a track in the body to facilitate guiding channel engager

down the track.

16. The method of claim 15, wherein releasing the interface includes: pushing a ramped distal surface of the one or more tabs onto a proximal end of the receiver assembly, thereby causing the distal end of the arm to be pushed outward and then fall into the one or more tool engagers of the receiver assembly.

17. The method of claim 15, further comprising: after adjusting the angle, advancing a channel engager of the adapter into a channel of the receiver assembly, thereby locking an angle of the receiver assembly with respect to the bone anchor.

18. The method of claim 15, further comprising: advancing the channel engager of the adapter toward a channel of the receiver assembly such that the channel engager contacts a set screw or rod disposed in the receiver assembly.

19. The method of claim 15, wherein the method is performed during a revision spinal surgery; and wherein the bone anchor was placed during a prior spinal surgery.

20. The method of claim 15, wherein the method further comprises: uncoupling the adapter from the receiver assembly; and connecting the receiver assembly with another receiver assembly via a rod.
